

Generalized linear models

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Last time: Exponential dispersion models

We say that a distribution belongs to the *exponential dispersion model* family if the distribution's probability function can be written in the form

$$f(y; \theta, \phi) = a(y, \phi) \exp \left\{ \frac{y\theta - \kappa(\theta)}{\phi} \right\}$$

Generalized linear models

For iid observations $(\mathbf{x}_1, Y_1), \dots, (\mathbf{x}_n, Y_n)$, a **generalized linear model** (GLM) consists of two components:

$$Y_i | \mathbf{x}_i \sim \text{EDM}(\mu_i, \phi) \quad \leftarrow \begin{aligned} &E[Y_i] = \mu_i \\ &\text{Var}(Y_i) = \phi v(\mu_i) \end{aligned}$$
$$g(\mu_i) = \underbrace{\mathbf{x}_i^T \boldsymbol{\beta}}_{\text{linear predictor}}$$

link function $\rightarrow$

μ_i and θ_i have a 1-to-1, increasing relationship.

often, we will choose g such that
 $\theta_i = g(\mu_i)$ (canonical link function)

Things we might do with a model

$$Y_i | \mathbf{x}_i \sim \text{EDM}(\mu_i, \phi)$$
$$g(\mu_i) = \mathbf{x}_i^T \boldsymbol{\beta}$$

- ▶ Estimate $\boldsymbol{\beta}$
- ▶ Model diagnostics
- ▶ Prediction
- ▶ Inference

Activity

Work on the activity (handout).